Does a "Fuzzy Soil Index" Help a Robot Walk on Soft Ground?

A replication study of the *wcci2026_hunter* fuzzy soil model

Hunter humanoid / IsaacSim / HumanoidVerse

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The problem (from the paper)

Goal: make a humanoid robot walk on **agricultural soil** — furrowed, deformable ground where the feet sink and slip.

- Real soil is **not uniform**: it changes from firm & grippy to soft & slippery.
- Two things vary independently:
  - **Traction** — how grippy the surface is (friction μ).
  - **Support** — how firm it is / how much the foot sinks (stiffness).
- The robot must **judge how hard the current soil is** to adapt and stay upright.

The paper's pitch

Combine traction + support into one human-readable "**soil difficulty score**" using **fuzzy logic**, and use it to drive training / reduce foot slip (>95% claimed).

What the "fuzzy soil index" actually is

A **fuzzy (Mamdani) controller**: two inputs $\rightarrow$ one difficulty score.

- Inputs: **traction** μ and **support** (stiffness).
- Each is graded *Low* / *Med* / *High* with **smooth** (fuzzy) boundaries.
- Rules $\rightarrow$ soil is *Easy* / *Moderate* / *Hard*.
- Output: one number $d \in [0.2, 0.8]$ (higher = harder).

Selling points: interpretable bins, smooth interpolation, blends two factors.

Two soil "dials"

traction μ & support (firmness)

$\downarrow$ fuzzy rules $\downarrow$

difficulty = Easy / Mod / Hard

Our question

Does the fuzzy index **actually add value** over just using the **raw soil numbers**?

- A fuzzy score is only useful if it **beats the raw measurements** it is built from.
- We replicate the recipe on the **Hunter** humanoid in **IsaacSim** (HumanoidVerse) and test the index four ways.
- We judge it as a **difficulty descriptor** and as a **control signal**.

Honest-negative friendly: we want the right answer, not a flattering one.

How we tested it — four settings

Study	**Simulator**	**Fuzzy used as**	**Support axis**
A	rigid PhysX	passive **label**	physically **inert**
B	rigid PhysX	**curriculum** controller	inert
C	**DFH deformable**	**descriptor**	physically **active**
① ②	DFH (+ noisy sensing)	descriptor / decisions	active

Each study closes the previous one's loophole. "Support axis" = whether soil *firmness* actually does anything in that simulator.

Sets A & B (rigid simulator)

Set A — fuzzy as a label: does *d* track difficulty / beat raw μ?

- *d* orders soil correctly (Spearman $\rho \approx 0.97$) ...
- ... but adds **nothing** over raw friction μ.
- **Why:** rigid PhysX has **no real "firmness" axis** — the foot can't sink, so the 2-dial model collapses to 1 dial (friction).

Set B — fuzzy drives the training curriculum:

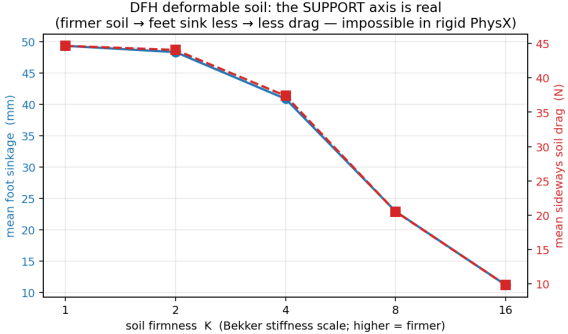
- fuzzy pacing $\approx$ crisp threshold $\approx$ fixed schedule (no significant difference).

Catch

A 2-input fuzzy model can't shine when **one input does nothing**. **We need a simulator where soil firmness is physically real.**

The fix: DFH deformable soil

DFH (Deformable Furrowed Heightfield) adds the missing physics: feet **sink** (Bekker), generate **sideways drag**, and the soil **deforms**.



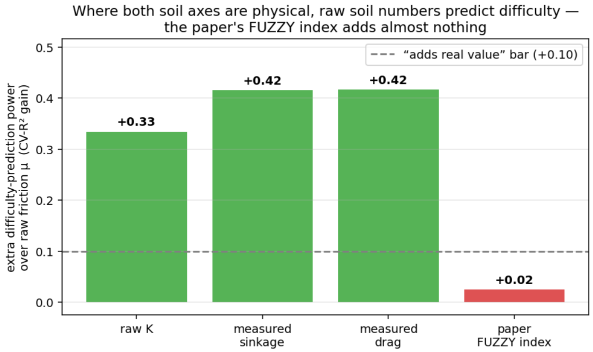
The support axis is now real:

- Soft soil (K=1): feet sink ∼**49 mm**.
- Firm soil (K=8): sink ∼**16 mm**; drag drops 33 → 9 N.
- Confirmed by a force-off control (no forces ⇒ no effect).

Same robot, same gait — only the soil firmness changes. You can see it.

Set C: the key result

Now **both** soil dials are physical. Does the fuzzy index finally add value?

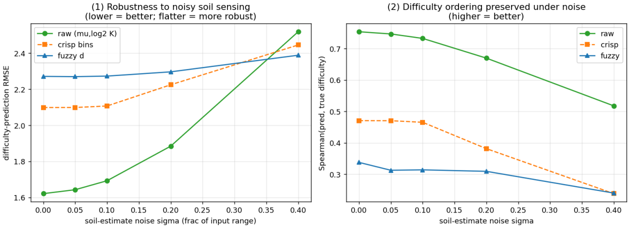


- Raw soil numbers **strongly predict** difficulty (+0.33 to +0.42).
- The paper's **fuzzy index** adds **+0.025** — essentially nothing (permutation p=0.001).

Sharp finding

Even where firmness **is** physically active, the fuzzy mapping **fails to use it**. The problem is the **fuzzy model's structure**, **not** the simulator.

Fuzzy logic's real strength is **tolerating imprecise inputs**. A real robot only *estimates* the soil. So we added sensing noise.



- Fuzzy **is** noise-stable (flat line) and overtakes raw *only* at extreme noise.
- But it wins by being **stable, not accurate** — raw still **orders soils better**.
- **Decisions** (②): a fuzzy-driven curriculum **camps on easy soil** and never advances.

Stability of an *uninformative* signal — it doesn't swing because it barely says anything.

Verdict

Bottom line

Across rigid **and** deformable soil, the paper's fuzzy soil index shows **no demonstrable value** over the raw soil numbers — as a label, a curriculum, a descriptor, or under noisy sensing.

- **Replicates:** the friction-curriculum **survival** benefit; DFH genuinely restores the firmness axis.
- **Does not add value:** the fuzzy mapping collapses two soil factors into a coarse score and **discards the useful information**.
- The failure is **structural**, not a simulator artifact (Set C proves it).

Scope & what would change the answer

Honest scope:

- This is about **this paper's specific fuzzy mapping** — *not* all 2-D or learned terrain descriptors.
- DFH is an **uncalibrated** deformable-soil surrogate; one frozen walker.
- Not a disproof of the paper on **real** soil — a non-replication *in simulation*.

What could make "fuzzy matter":

- Redesign the fuzzy rules so they **actually use the firmness axis** (they ignore it at mid-traction), or use a **learned** 2-D descriptor.
- Train a soil-robust walker (removes the single-policy caveat).

A fuzzy score is only useful if it beats the numbers it's made from.

Here, it doesn't.

Code & full write-ups: `docs/experiments/fuzzy_soil_UNIFIED_conclusion.md`; models on Hugging Face `anhrisn/hunter-dfh-locomotion`.